

Mark schemes

Q1.

(a) (i) Poisson

B1

1

(ii) $E(3X - 1) = 3\lambda - 1$

B1

$\text{Var}(3X - 1) = 9\lambda$
oe (allow $3^2\lambda$)

B1

2

(iii) $P(X = x + 1) = \frac{e^{-\lambda} \times \lambda^{x+1}}{(x+1)!}$

B1

$P(X = x + 1) = \frac{e^{-\lambda} \times \lambda^{x+1}}{(x+1)!}$

$= \frac{e^{-\lambda} \times \lambda^x \times \lambda}{(x+1)x!}$
 $= \frac{\lambda}{x+1} \times \frac{e^{-\lambda} \times \lambda^x}{x!}$

dep(B1)

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$= \frac{\lambda}{x+1} P(X = x)$

AG

Adep1

3

(b) (i) $\lambda_{\text{car}} = 500$ / hour

$\lambda_{\text{coach}} = 10$ / hour

$\Rightarrow \lambda_{\text{vehicle}} = 510$ / hour = 8.5 / min
for 8.5 stated / used

B1

$P(V \geq 10) = 1 - 0.6530$

M1

$$= 0.347$$

A1

special case:

$$\lambda = 10 \Rightarrow B1M0A0$$

$$B1 \Rightarrow 1 - 0.458 \text{ or } 0.542$$

3

(ii) $\mu_{\text{car}} = 836/\text{hour}$

$$\mu_{\text{coach}} = 22/\text{hour}$$

$$\Rightarrow \mu_{\text{vehicle}} = 858 / \text{hour} = 14.3 / \text{min}$$

for 14.3 stated / used

B1

$$P(V \leq 3) = P(V = 0, 1, 2, 3)$$

$$\begin{cases} e^{-14.3} \left[1 + \frac{14.3}{1} + \frac{14.3^2}{2} + \frac{14.3^3}{6} \right] \\ e^{-14.3} \times 604.91283 \end{cases}$$

$$= 0.0003726 \text{ to } 0.000373$$

*All 4 terms required for **any** $\lambda > 0$*

M0 for use of normal approximation

M1

$$= 0.00037 \text{ (2sf)}$$

dep M1

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Adep1

3

[12]

Q2.

$$X \sim \text{Po}(\lambda)$$

(a) $E(Z) = 4 \times 2.5 + 30 = 40$

CAO

B1

$$\text{Var}(Z) = 4^2 \times 2.5$$

Use of $V(aX) = a^2 V(X)$

Ignore '+30'

M1

$$= 40 \{= E(Z)\}$$

CAO AG

A1

Note:

$$4 \times 2.5 + 30 = 4^2 \times 2.5 \Rightarrow \text{B1 M1 A0}$$

$$\text{plus value of 40 quoted} \Rightarrow \text{B1 M1 A1}$$

3

- (b) No values less than 30 are possible
 No odd values are possible
 Only even values are possible
 Only values of 30, 34, 38, etc, are possible
Or equivalent

B1

1

[4]

Q3.

- (a) (i) $X \sim \text{Po}(5.0)$
 $\Rightarrow P(X < 4) = P(X \leq 3)$
(0.440 to 0.441) for B1
 $= 0.265$
 CAO

B2

2

- (ii) $Y \sim \text{Po}(1.5)$
 $\Rightarrow P(Y = 4) = \frac{e^{-1.5} \times (1.5)^4}{4!}$

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M1

$$= 0.0471$$

(0.047 to 0.0471)

A1

2

- (b) (i) $T = X + Y \sim \text{Po}(6.5)$

B1

$$\Rightarrow P(T > 5) = 1 - P(T \leq 5)$$

(1 - 0.2237) or (1 - 0.5265)

B1

$$= 1 - 0.369$$

$$= 0.631$$

B1

3

- (ii) $p = {}^8C_7 (0.631)^7 (0.369) + (0.631)^8$
ft on their p from (b)(i)
Either part attempted

M1ft

$$p = 0.11758 + 0.02513$$

(both parts correct)

A1ft

$$= 0.143$$

AWFW 1.142 to 0.143 (CAO)

A1

3

- (c) (i) Mean = 8
CAO
- Variance = $s^2 = 16.9$
(sample variance = 15.2)
(AWRT)

B1

B1

2

- (ii) Poisson not a good model for data

B1dep

Mean $\neq$ Variance

B1

2

[14]

Q4.

- (a) (i) $X \sim \text{Po}(9 \ 0) \Rightarrow$
 standard deviation = 3

B1

1

- (ii) $P(6 < X < 12)$
 $= P(X \leq 11) - P(X \leq 6)$

M1

$$= 0.8030 - 0.2068$$

M1ft

$$= 0.5962$$

CAO

A1

3

(b) (i) $T \sim \text{Po}(11.5)$

CAO

B1

1

(ii) $P(T \leq 1) = P(T = 0) + P(T = 1)$
 $= e^{-11.5} + 11.5e^{-11.5}$
 $= 0.000127$

Use of $T = 0$ and 1

M1

Substitute correctly into formula

M1

AWFW 0.000126 and 0.00013

A1

3

(c) $\bar{x} = 12.0$ and $s^2 = 19.3$ ($s = 4.40$)
 $\sigma^2 = 17.4$ ($\sigma = 4.17$)

B1

Mean and variance very different

dep on s^2 (or σ^2)

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E1

$\Rightarrow \text{Po}(12.0)$ not a suitable model
 (dep E1)

E1

3

[11]

Q5.

(a) (i) $P(A = 4) = \frac{e^{-3.5} \times (3.5)^4}{4!} = 0.189$

M1A1

2

(ii) $P(B \leq 6) = 0.762$

B1

1

(iii) $T = A + B \sim \text{Po}(8.5)$

$P(T \text{ fewer than } 10) = P(T < 10)$

Use of Po(8.5)

M1

$= P(T \leq 9)$

T ≤ 9 attempted

M1

$= 0.653$

CAO

A1

3

(b) $X \sim B(5, 0.653)$

X ~ B(5, their p)

B1

$P(X \geq 4) = \binom{5}{4} (0.653)^4 (0.347) + (0.653)^5$

M1

$= 0.31547 + 0.11873$

$= 0.434$

On their p from (a)(iii)

A1ft

3

(c) (i) $\bar{x} = 9.2$

B1

$s^2 = 9.29$

$\sigma^2 = 8.36$

B1

2

- (ii) Mean and variance have similar values which suggests that Poisson distribution may be appropriate

Q6.

(a) (i) $P(X \leq 1) = 0.9098$

M1

AWRT 0.91

A1

2

(ii) $P(Y = 2) = \frac{e^{-2.5} \times (2.5)^2}{2!}$

M1

$= 0.257$

AWFW 0.256 and 0.257

A1

2

(b) (i) $T = X + Y \sim P_o(3.0)$

$P(\text{Total of 2 breakdowns}) = p([x,y] = 2)$

$= P([2,0] \text{ or } [0,2] \text{ or } [1,1])$

B1

$P(T = 2) = \frac{e^{-3.0} \times (3.0)^2}{2!}$

$= [0.0758 \times 0.0821] + [0.6065 \times 0.25652]$

$+ [0.20521 \times 0.3033] = 0.006223 + 0.155585$

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M1

$= 0.224$

$+ 0.062233$

$= 0.224$

A1

3

OR

Using tables: $= 0.4232 - 0.1991$

$= 0.224$

(ii) $p = (0.224)^4 = 0.0025$

M1 A1

2

(c) (i) $4X \sim P_o(2.0) \Rightarrow \mu = 12.0$

$$4Y \sim P_o(10.0)$$

B1

1

$$(ii) \quad P(T \geq 18) = 1 - P(T \leq 17)$$

M1

$$= 1 - 0.9370$$

$$= 0.0630$$

A1

2

[12]

Q7.

$$(a) \quad (i) \quad P(2 \text{ or fewer}) = 0.9526$$

$$0.953 \text{ (} 0.952 \sim 0.953 \text{)}$$

B1

$$(ii) \quad P(>1, <3) = P(2)$$

$$P(2) \text{ required}$$

M1

$$= P(2 \text{ or fewer}) - P(1 \text{ or fewer})$$

$$P(2 \text{ or fewer}) - P(1 \text{ or fewer}) \text{ or correct use of formula}$$

m1

$$= 0.9526 - 0.8088 = 0.144$$

$$0.144 \text{ (} 0.143 \sim 0.1445 \text{)}$$

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A1

4

$$(b) \quad (i) \quad \text{mean } 5 \times 0.8 = 4$$

$$4 \text{ cao}$$

B1

$$\text{standard deviation } \sqrt{4} = 2$$

M1

$\sqrt{\text{their mean} - \text{allow variance} = \text{their mean} - \text{variance must be stated}}$
2 cao

A1

3

$$(ii) \quad P(\text{fewer than } 3) = 0.2381$$

$$(0.2375 - 0.2385)$$

B1

1

[8]

Q8.

- (a) (i) $P(3) = 0.9212 - 0.7834 = 0.138$
*P (3) = P (3 or fewer) – P (2 or fewer) or
 use of correct formula*

M1

0.138 (0.1375 to 0.1385)

A1

2

- (ii) Poisson mean 2.4
Poisson, mean 2.4

B1

$P(0) = 0.0907$
0.0907 (0.0907 to 0.09075)

B1

2

- (iii) Poisson mean 12
Poisson mean 12

B1

$P(20 \text{ or more}) = 1 - P(19 \text{ or fewer})$
P (20 or more) 1 – P (19 or fewer)

M1

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$= 1 - 0.9787$
 $= 0.0213$

A1

3

0.0213 (0.021 to 0.0214)
sc allow B2 for 0.0116 (0.011 to 0.0117)

- (b) (i) Poisson mean 1.8
 Standard deviation $\sqrt{1.8} = 1.34$
ft their mean

M1

1.34 (1.34 to 1.345)

A1

2

- (ii) Cannot distinguish between $2 - 1$ and $3 - 0$

Reason

E1
1

(c) Mean not constant

Reason - generous

E1
1

[11]

Q9.

(a) (i) $P(6 \text{ or fewer}) = 0.3782$

Correct use of Poisson tables or formula

0.378 (0.378 – 0.379)

M1 A1

(ii) $P(8) = 0.6620 - 0.5246 = 0.137$

$P(8) = P(8 \text{ or fewer}) - P(7 \text{ or fewer})$

or use of correct formula

0.137 (0.137, 0.138)

M1 A1

4

(b) $\sqrt{7.5} = 2.74$

method

sc B1 var = 7.5

2.74(2.73, 2.74)

M1 A1

2

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[6]

Q10.

(a) Poisson

Poisson

B1
1

(b) (i) $x = 3.69$

$s^2 = 7.06$, allow 6.52

B1

3.69 (3.68 – 3.7)

B1

2

7.06 (7.06 – 7.07) or 6.52 (6.51 – 0 6.53)

(ii) No, mean and variance very different

M1

Attempted comparison of mean and variance

A1

2

*No - must be clearly based on variance
e.g. disallow if compared with s.d. even if
they call it variance*

[5]



EXAM PAPERS PRACTICE

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Examiner reports

Q1.

In part (a)(i), the vast majority of candidates correctly stated the distribution of X to be 'Poisson'. In part (a)(ii), although most stated the answers correctly as $E(3X - 1) = 3\lambda$ and $\text{Var}(3X - 1) = 9\lambda$ some, who seemed to understand the underlying concepts by stating $E(3X - 1) = 3 \times E(X) - 1$ and $\text{Var}(3X - 1) = 3^2 \times \text{Var}(X)$, then failed to state their answers in terms of λ as requested, consequently gaining no marks. Part (a)(iii) proved to be very

popular with most candidates. They were able to correctly state $P(X = x + 1) = \frac{e^{-\lambda} \times \lambda^{x+1}}{(x+1)!}$ and then were able to use

$\lambda^{x+1} = \lambda \times \lambda^x$ and $(x + 1)! = (x + 1) \times x!$ to prove the given answer.

In part (b), it was very pleasing to see that candidates dealt with such phrases as 'at least 10' and 'at most 3' much better than in previous series. In part (b)(i), $\lambda = 8.5$ followed by a correct calculation was often seen. In part (b)(ii), $\lambda = 14.3$ was correctly seen and often used to find the correct answer of 0.00037. Unfortunately, some failed to comply with the instruction 'give your answer to two significant figures' and consequently lost a mark. Those who chose to do the whole of this part of the question on their calculators either gained full marks when the correct answer was stated or lost at least two marks for incorrect answers with no method shown.

Q2.

Most candidates provided a convincing answer in part (b). Also, it was very rare to see an incorrect answer to part (a) but even rarer to see a correct reason in part (b), where the most common answer was ' λ too large', which, needless to say, did not score the mark.

Q3.

There were many fully-correct solutions seen to this question. The vast majority of candidates used the tables provided to enable them to find the correct answer to part (a)(i) by the most efficient method. A minority of candidates seemed only able to use the formula and consequently worked out the individual probabilities before obtaining their answer. In part (a)(ii), the formula was often used correctly to obtain the correct answer of 0.0471. However, there were some candidates who seemed to think that they had to use tables whatever the value of λ and so attempted to average the values found in the tables under $\lambda = 1.4$ and $\lambda = 1.6$ incorrectly thinking that this would give them the required value for $\lambda = 1.5$.

In part (b)(i), most candidates interpreted 'more than 5' correctly using $P(X + Y > 5) = 1 - P(X + Y \leq 5)$, whilst others incorrectly used $1 - P(X + Y \leq 4)$ or $1 - P(X + Y \leq 6)$. Only the most able candidates then went on to use their answer to correctly determine the numerical solution to part (b)(ii). There were many erroneous attempts at using various expressions with $0.631^7 + 0.631^8$ or $0.631^7 \times 0.369$ or even $6.5 \times 0.631^7 \times 0.369$ seen, rather than the required $p = {}^8C_7(0.631)^7(0.369) + (0.631)^8$. In part (c)(i), the vast majority of candidates correctly stated that $\bar{x} = 8$ and that $s^2 = 16.9$, but with some stating the sample variance as 15.2. These answers usually enabled candidates to indicate correctly in part (c)(ii) that a Poisson distribution was probably not a good model since the mean

and the variance had very different values.

Q4.

In part (a)(i), there were surprisingly many candidates who did not know the relationship between the value of λ and the standard deviation of a Poisson distribution. In part (a)(ii), although the vast majority realised that the most efficient method of answering this part of the question required the use of the table on page 23 of the formulae booklet, very many failed to look up the appropriate values to enable them to show that $P(6 < X < 12) = P(X \leq 11) - P(X \leq 6) = 0.596$. Those candidates who calculated the separate probabilities for $X = 7, 8, 9, 10$ and 11 often reached the required answer.

In part (b)(i), it should be noted that $\lambda = 11.5$ and an indication that the distribution of T is Poisson were both required to gain the mark. In part (b)(ii), most candidates realised that “fewer than 2” implied that $P(T \leq 1) = P(T = 0) + P(T = 1)$. The correct answer of 0.000127

was then usually found by the use of $\lambda = 11.5$ and the formula $P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$ (found on page 11 of the formulae booklet).

Although part (c) asked candidates to calculate appropriate numerical measures (plural), this was ignored by some candidates. Such candidates usually calculated $\bar{x}$ and, finding this to coincide with the given value of λ , stated inappropriately that Po(12) was a suitable model for the given data. Of those candidates who did calculate more than one numerical measure, most correctly calculated the mean and variance of the given data ($\bar{x} = 12$; $s^2 = 19.3$ or $\sigma^2 = 17.4$), and then indicated that, because the mean and variance were (very) different values, Po(12) was not a suitable model for the number of places booked each week. Some candidates incorrectly thought that the mean and the standard deviation had to be the same.

Q5.

Part (a) was generally answered well. However, there were some candidates who attempted to use tables in part (a)(i), even though values are not tabulated for $\lambda = 3.5$, whilst in part (a)(iii) some candidates could not interpret ‘fewer than 10’ to mean ‘ ≤ 9 ’. It was, however, very pleasing to see fewer candidates than usual attempting to use the formula in part (a)(ii). Only the most able candidates managed to complete part (b) successfully, with the majority not realising that there were 5 ways of selecting 4 days from 5.

Part (c) was done really well by most candidates, although, in part (c)(ii), several thought that the mean had to be equal to the variance in order for a Poisson distribution to form a suitable model. Here, $\bar{x} = 9.2$ and $s^2 = 9.29$ were close enough to each other to draw this conclusion.

Q6.

Answers to part (a)(i), were invariably correct with most candidates using the statistical tables provided. A minority of candidates calculated $P(X = 0) + P(X = 1)$ using the correct probability function for Po(0.5) and were often successful. Part (a)(ii) was done well by all but the weaker candidates who usually attempted to use tables to find $P(X = 2)$ for $\lambda = 2.4$ and $\lambda = 2.6$ and then find the mean of the two values obtained! In part (b)(i), most candidates considered Po(3.0) and then used the formula or tables to achieve correctly the given answer of 0.224. In part (b)(ii) most candidates correctly considered $(0.224)^4$ and so scored the 2 marks available. The majority of candidates, in part(c)(ii), correctly

interpreted ' T at least 18' as meaning ' $1 - (T \leq 17)$ '.

Q7.

Nearly all candidates were able to use Poisson tables but fewer realised that the probability of 'more than one but fewer than three' is $P(2)$. Part (b)(i) was generally well answered with fewer than expected confusing standard deviation and variance.

Q8.

This was generally well answered. Some confused standard deviation with variance in part (b)(i).

Q9.

Most candidates succeeded with part (a). In part (b), the most common incorrect answers were 7.5 or 7.52.

Q10.

Confusion between variance and standard deviation was not uncommon. Only candidates who realised they were supposed to compare the mean and variance of the given data made progress in part (b)(ii).